Question 1: Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

Course outcome: CO1, CO2

The Lab 1 Flask application was containerized: a Dockerfile was written, the image built, the container run and verified, and the image layers and container behavior documented.

Dockerfile

```
# Lab 2 - container image for the flask app
FROM python:3.12-slim

ARG APP_VERSION=2.0.0
ENV APP_VERSION=${APP_VERSION}

WORKDIR /app

# install dependencies first so this layer is cached
COPY app/requirements.txt /app/requirements.txt
RUN pip install --no-cache-dir -r /app/requirements.txt

# copy application code
COPY app/ /app/

EXPOSE 5000

# run as non-root user
RUN useradd -m appuser
USER appuser

CMD ["python", "app.py"]
```

Build and run

```
$ docker build -f docker/Dockerfile -t devops-lab2-app:2.0.0 .
...
$ docker images | grep devops
devops-lab2-app   2.0.0   66c210e3f934   204MB

$ docker run -d --name lab2-app -p 5000:5000 devops-lab2-app:2.0.0
$ docker ps --filter name=lab2-app
NAMES      IMAGE                  STATUS         PORTS
lab2-app   devops-lab2-app:2.0.0  Up             0.0.0.0:5000->5000/tcp
```

Container behavior

```
$ curl http://localhost:5000/health
{"status":"UP","version":"2.0.0"}

$ docker top lab2-app
UID      PID    CMD
abhinav  4270   python app.py

$ docker exec lab2-app python --version
Python 3.12.16
```

The `docker top` UID shows the container process running as uid 1000, which is the non-root `appuser` created in the Dockerfile mapped to the host.

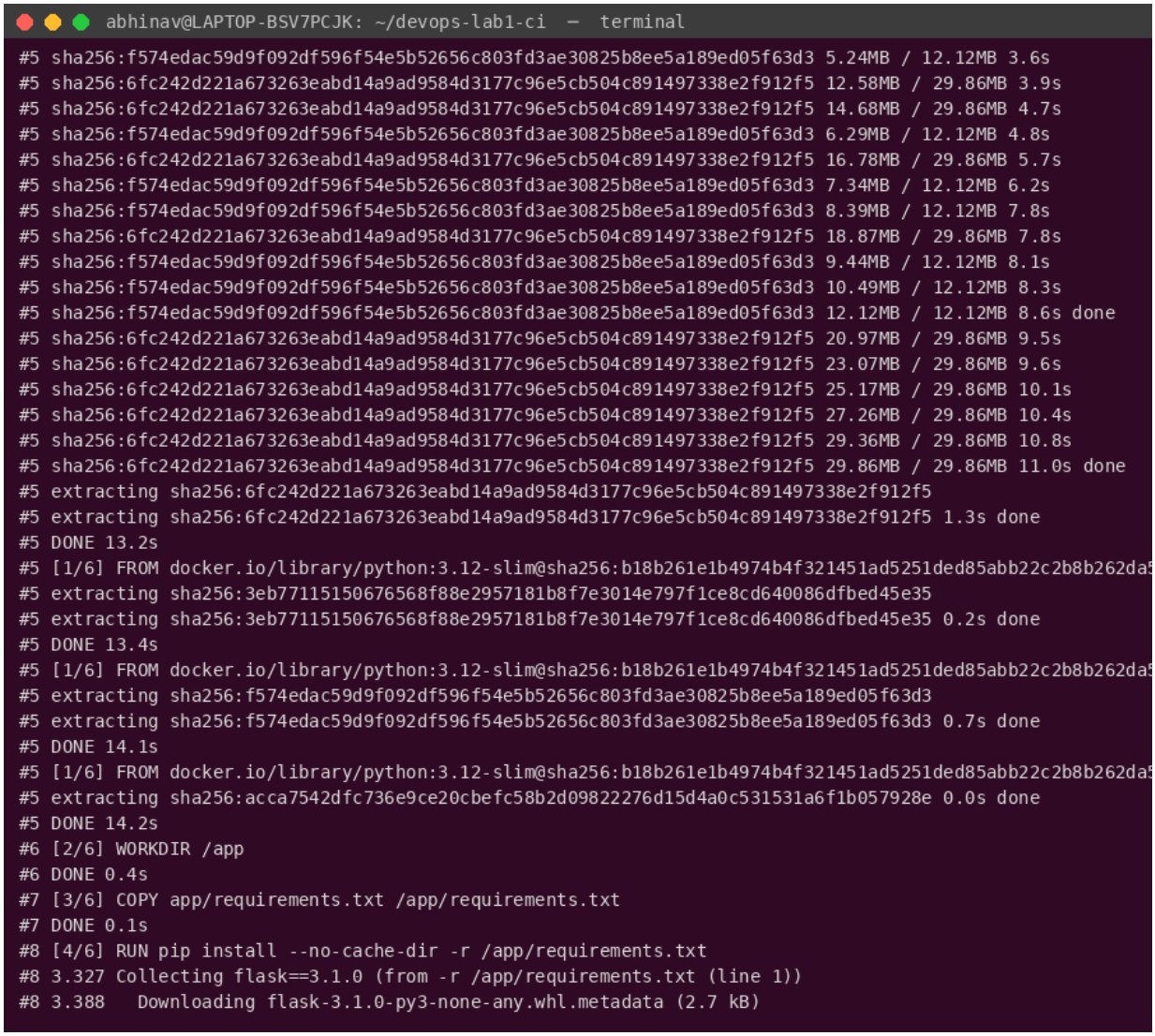


Fig: docker build output (layers) and docker images

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ docker history devops-lab2-app:2.0.0
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
66c210e3f934   11 seconds ago   CMD ["python" "app.py"]                         0B        buildkit.dockerf
<missing>      11 seconds ago   USER appuser                                    0B        buildkit.dockerf
<missing>      11 seconds ago   RUN /bin/sh -c useradd -m appuser # buildkit    69.6kB    buildkit.dockerf
<missing>      11 seconds ago   EXPOSE [5000/tcp]                               0B        buildkit.dockerf
<missing>      12 seconds ago   COPY app/ /app/ # buildkit                      20.5kB    buildkit.dockerf
<missing>      12 seconds ago   RUN /bin/sh -c pip install --no-cache-dir -r…   19.9MB    buildkit.dockerf
<missing>      18 seconds ago   COPY app/requirements.txt /app/requirements.…   12.3kB    buildkit.dockerf
<missing>      18 seconds ago   WORKDIR /app                                    8.19kB    buildkit.dockerf
<missing>      6 days ago       CMD ["python3"]                                 0B        buildkit.dockerf
<missing>      6 days ago       RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerf
<missing>      6 days ago       RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.4MB    buildkit.dockerf
<missing>      6 days ago       ENV PYTHON_SHA256=b5796915dc45ff745cfb4e990a…   0B        buildkit.dockerf
<missing>      6 days ago       ENV PYTHON_VERSION=3.12.16                      0B        buildkit.dockerf
<missing>      6 days ago       ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerf
<missing>      6 days ago       RUN /bin/sh -c set -eux;  apt-get update;  a…   4.93MB    buildkit.dockerf
<missing>      6 days ago       ENV LANG=C.UTF-8                                0B        buildkit.dockerf
<missing>      6 days ago       ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B        buildkit.dockerf
<missing>      7 days ago       # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.7MB    debuerreotype 0.
abhinav@LAPTOP-BSV7PCJK:~$ docker inspect lab2-app --format "IP={{.NetworkSettings.IPAddress}} Ports={{.Ne
template parsing error: template: :1:21: executing "" at <.NetworkSettings.IPAddress>: map has no entry fo
abhinav@LAPTOP-BSV7PCJK:~$ docker inspect devops-lab2-app:2.0.0 --format "Layers={{len .RootFS.Layers}} Dr
template parsing error: template: :1:51: executing "" at <.GraphDriver.Name>: map has no entry for key "Gr
abhinav@LAPTOP-BSV7PCJK:~$ docker logs lab2-app
 * Serving Flask app 'app'
 * Debug mode: off
[31m[1mWARNING: This is a development server. Do not use it in a production deployment. Use a production
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
[33mPress CTRL+C to quit[0m
abhinav@LAPTOP-BSV7PCJK:~$ docker exec lab2-app python --version
Python 3.12.16
abhinav@LAPTOP-BSV7PCJK:~$ docker top lab2-app
UID                 PID                 PPID                C                   STIME               TTY
abhinav             4270                4234                31                  19:48               ?
```

Fig: docker history - image layers, inspect, logs and top



Fig: Containerized application served on port 5000

Image layer documentation

- Base layer: python:3.12-slim (bulk of the 204 MB).
- Layer 2: COPY requirements.txt - only changes when dependencies change.
- Layer 3: pip install - cached on rebuilds because it sits above the requirements copy.
- Layer 4: COPY app/ - invalidated on every code change (small, fast).
- Layer 5: useradd + USER - runtime security, non-root process.
- Total: 8 layers, 204 MB uncompressed (49.6 MB compressed on Docker Hub).

Observations

- Placing the dependency install before the code copy makes rebuilds fast (layer caching).
- The Flask dev server binds 0.0.0.0 so it is reachable outside the container.
- The container responds within a second of `docker run -d`; `docker logs` shows the Flask banner.
- Docker 29.x on WSL2 defaults to the systemd cgroup driver; `native.cgroupdriver=cgroupfs` was set in /etc/docker/daemon.json because the minikube docker driver fails with the systemd driver.

Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Docker, Docker Compose, Minikube and kubectl. The evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-2 under the docs/ directory.